Marvell Pushes Shared Optical Memory Across AI Racks for KV Cache

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Marvell is extending the AI memory hierarchy beyond individual servers and racks with a Photonic Fabric Memory Appliance (PFMA) designed to make tens of terabytes of shared DDR5 memory accessible to multiple AI processors over optical links. Speaking at Hot Interconnects on August 20, Ravi Mahatme, Senior Director of Product Management in Marvell's Photonic Fabric Business Unit, presented the architecture as a new memory tier for large-scale AI inference, particularly for key-value (KV) cache. A single rack-mountable PFMA is designed to connect as many as 16 servers or XPUs across multiple racks, providing up to 1 TB of HBM3e plus 128 standard DDR5 DIMM slots supporting approximately 16 TB to 32 TB of capacity in a unified shared address space.

The architecture targets a growing inference bottleneck: KV cache expands with model size, context length, batch size and the number of concurrent conversations. HBM provides high bandwidth but limited capacity and must also accommodate model weights and activations, while host DRAM has greater capacity but remains constrained at the server level. SSD and network-attached storage can provide much larger capacity but at substantially higher latency. Marvell's approach inserts an optically connected shared-memory tier between local memory and storage. Mahatme said this could allow KV cache generated by one GPU or server to be accessed by another processor without first moving the cache across the conventional front-end packet network. This creates a cross-node KV cache that can also give inference schedulers greater freedom to assign prefill and decode workloads across available accelerators.

The PFMA is built from 16 Photonic Fabric Memory Modules connected through a passive all-to-all fiber shuffle. Each module combines 72 GB of HBM3e, interfaces for as many as eight DDR5 DIMMs, a Photonic Fabric ASIC and optical I/O delivering 7.2 Tbps. HBM operates as a write-through cache for the DDR memory, while hardware semaphores provide mechanisms for software coherency. Marvell outlined two host connectivity models: a PCIe Gen6/CXL 3.1 PF-NIC providing 2 Tbps of bandwidth and roughly 300 ns average loaded access latency over one meter of fiber, or a Photonic Fabric chiplet integrated directly with an XPU package supporting more than 16 Tbps and approximately 230 ns average latency. The company says the underlying optical technology is qualified for reach up to 50 meters, although Mahatme said approximately 25-30 meters is a more appropriate design target for shared memory because latency increases with fiber distance.

Marvell Photonic Fabric Memory Appliance — Key Architecture Points

Pod-Scale Optical Memory: One PFMA can serve up to 16 servers, XPUs, or a combination across multiple racks.
Memory Capacity: Up to 1 TB HBM3e + 128 DDR5 DIMM slots, providing approximately 16 TB or 32 TB depending on DIMM capacity.
Shared Address Space: Connected processors access a unified shared memory pool rather than maintaining isolated capacity at each server.

Memory Module: Each Photonic Fabric Memory Module integrates 72 GB HBM3e and supports up to eight DDR5 DIMMs.
Optical Bandwidth: Each memory module provides 7.2 Tbps optical I/O using 16 transmit and 16 receive lanes at 224 Gbps.
PF-NIC: PCIe Gen6 x16 / CXL 3.1 server interface delivering 2 Tbps, with approximately 300 ns average loaded latency over one meter of fiber.
PF-Chiplet: Package-level XPU integration delivering more than 16 Tbps and approximately 230 ns average loaded latency.
Optical Reach: Photonic Fabric technology is qualified for approximately 50 meters; Marvell sees roughly 25-30 meters as appropriate for the memory use case.
Fiber Fabric: A passive all-to-all fiber shuffle connects front-panel ports with the appliance's 16 memory modules.
External Lasers: 32 external laser sources: 16 illuminate the local memory modules and 16 support remote PF-NICs or PF-Chiplets.
KV Cache Sharing: Processors attached to the shared tier can access KV cache generated by other processors, enabling cross-node KV cache and greater scheduling flexibility.
Dynamic Allocation: CXL 3.1 dynamic capacity allocation allows memory to be requested and released during operation.

In simulation using Llama-405B and compute traces derived from an eight-H200 system, Marvell compared a conventional configuration with 2 TB of CPU DRAM against a PFMA configuration providing 32 TB. At 50 simultaneous conversations, both configurations produced similar time-to-first-token (TTFT), but the CPU-memory configuration degraded sharply as concurrency and KV cache requirements increased. The PFMA simulation maintained relatively flat TTFT through 300 conversations and an approximately 82% KV cache hit rate. Marvell characterized this as more than a 6X concurrency improvement - roughly 300 conversations versus 50 - at approximately the same TTFT. These results are simulation-based rather than measurements from production silicon; the presentation specifically identifies the latency figures as pre-silicon numbers subject to change. 'The biggest value from having a shared memory tier is you can have a cross-node KV cache,' Mahatme said.

Analysis

Marvell is effectively proposing that memory become another fabric-connected resource in an AI cluster. CXL began that transition by separating memory capacity from individual CPUs at server and rack scale. Photonic Fabric pushes the model farther by using optical reach to make memory accessible across multiple racks while attempting to retain hundreds-of-nanoseconds rather than storage-class latency. The important architectural shift is therefore not simply 32 TB of additional DRAM; it is the possibility of treating KV cache as cluster infrastructure rather than data belonging to an individual accelerator.

The KV cache use case also illustrates why AI inference architecture is increasingly about data movement as much as compute. Disaggregated prefill and decode can improve accelerator utilization, but separating those stages creates another problem: the KV cache generated during prefill must reach the processor performing decode. A shared memory tier potentially changes

that operation from explicitly transferring a large data structure between compute nodes to allowing both processors to access the same memory resource. Whether this translates into production gains will depend on realized silicon latency, bandwidth under contention, coherency and software integration, but Marvell's simulations establish the performance target the architecture is intended to address.

Photonic Fabric entered Marvell through its February 2, 2026 acquisition of Celestial AI. The technology now sits alongside Marvell's Structera CXL portfolio, custom XPU silicon, Ethernet switching and optical interconnect portfolio. The PFMA illustrates how Marvell can apply the acquired optical technology beyond XPU-to-XPU scale-up links: the same underlying concept of moving high-bandwidth connections from electrical traces onto optics can connect compute directly to large shared memory pools. In that sense, Photonic Fabric is becoming a platform spanning both scale-up connectivity and memory disaggregation rather than a single-purpose optical interconnect product.